Readout by temporarily enlarging a stabilized cat

DuQuantum 2026 Alice & Bob challenge: Task 2.4(a-d)

Computational research report

25 October 2026

The goal is to read the memory bit faster while retaining its protection against unwanted switching. We test **enlarge, measure, restore**: increase the cat amplitude from 2.1 to 2.6, apply a stronger memory drive, then return to storage. In the ideal two-mode model, the complete cycle reaches the same calculated signal target in **47.40 µs instead of 110.22 µs** for the specified weak reference. Its estimated readout-stage relaxation rate is below undriven storage, and its final wrong-state population is smaller than the weak reference's.

Task 2.4(a): derive the readout Hamiltonian

Pumped circuit and two-photon exchange

With $\hbar = 1$, the circuit Hamiltonian is

$$H_{\text{circuit}} = \omega_a a^\dagger a + \omega_b b^\dagger b - 2E_J \cos\varphi_\Sigma \cos(\hat{\varphi} + \varphi_\Delta), \qquad \hat{\varphi} = \varphi_a(a + a^\dagger) + \varphi_b(b + b^\dagger).$$

The operators a and b annihilate memory and buffer photons; φ_a and φ_b are their zero-point phases. At $\varphi_\Delta = \pi/2$ and $\varphi_\Sigma = -\pi/2 + \epsilon_p \cos\omega_p t$, the identities $\cos(\hat{\varphi} + \pi/2) = -\sin\hat{\varphi}$ and $\cos\varphi_\Sigma \simeq \epsilon_p \cos\omega_p t$ give

$$H_{\text{ATS}}(t) \simeq 2E_J\epsilon_p \cos\omega_p t \left(\hat{\varphi} - \frac{\hat{\varphi}^3}{6} + \frac{\hat{\varphi}^5}{120} + \cdots\right).$$

The mixed cubic contribution is

$$H^{(3)}_{\text{ATS}} \supset -E_J\epsilon_p\varphi_a^2\varphi_b \cos\omega_p t (a + a^\dagger)^2(b + b^\dagger).$$

Here $(a + a^\dagger)^2 = a^2 + a^{\dagger 2} + 2a^\dagger a + 1$. In the rotating frame, $a^2 b^\dagger$ acquires $e^{-i(2\omega_a - \omega_b)t}$. Choosing $\omega_p = 2\omega_a - \omega_b$ cancels that rotation through one pump harmonic. The factor $1/2$ from the cosine gives

$$H_{\text{ex}} = g_2(a^2 b^\dagger + a^{\dagger 2} b), \qquad |g_2| = \frac{E_J\epsilon_p\varphi_a^2\varphi_b}{2}.$$

This pump phase gives negative exchange. We shift it by π and use $g_2 > 0$ throughout. For $(\omega_a, \omega_b, \omega_p)/2\pi = (4.0, 4.8, 3.2)$ GHz, $E_J/2\pi = 40$ GHz, $\varphi_a = 0.06$ and $\varphi_b = 0.30$, the rounded $\epsilon_p = 0.035$ gives $|g_2|/2\pi = 0.756$ MHz. Using $\epsilon_p = 0.0353241$ gives the specified 0.763 MHz. These are illustrative circuit parameters [1,3].

Separate microwave drives

The flux pump creates the exchange interaction. A separate resonant buffer microwave supplies $H_d = -\epsilon_d(b + b^\dagger)$ and sets $\alpha^2 = \epsilon_d/g_2$. A further memory microwave port supplies the readout force. With the explicit laboratory phase convention

$$H_{Z,\mathrm{lab}} = 2\epsilon_Z(t)\cos(\omega_a t - \phi)(a + a^\dagger),$$

the resonant rotating-frame term is $H_Z = \epsilon_Z(t)(e^{i\phi}a^\dagger + e^{-i\phi}a)$. The discarded terms rotate at $2\omega_a$. A charge-coupled port realizes the same convention after a carrier-phase choice. All main calculations use $\phi = 0$, giving

$$\boxed{H = g_2(a^2b^\dagger + a^{\dagger 2}b) - \epsilon_d(b + b^\dagger) + \epsilon_Z(t)(a + a^\dagger).}$$

The memory drive is therefore an explicitly added physical control, rather than an automatic consequence of the specified ATS pump.

Corrections to the leading interaction

The first-order linear terms are off resonance at the supplied frequencies. Keeping the resonant exchange through fifth order gives

$$\begin{aligned} H_{\mathrm{ex}}^{(\le 5)} = {} & g_2[1 - (\varphi_a^2 + \varphi_b^2)/2]a^2b^\dagger \\ & - \frac{g_2\varphi_a^2}{3}a^\dagger a^3 b^\dagger - \frac{g_2\varphi_b^2}{2}a^2 b^{\dagger 2} b + \mathrm{h.c.} \end{aligned}$$

The last two terms still exchange two memory photons for one buffer photon, but their amplitudes depend on occupation. The first term includes the associated correction to the cubic coefficient. An odd sine potential gives no static Kerr term at first pump order. Off-resonant processes can generate frequency shifts and Kerr interactions at second pump order; those are effective corrections, not terms retained directly by the first-order rotating-wave approximation. Appendix A derives and tests these corrections. The numerical protocol below uses the leading cubic model: the buffer drive controls cat size, and the memory drive produces the readout signal.

Task 2.4(b): establish the contrast-error trade-off

Model and weak-drive response

We solve both modes, with time in µs and angular frequencies in $\mu\mathrm{s}^{-1}$:

$$\dot\rho = -i[H, \rho] + \kappa_a\mathcal{D}[a]\rho + \kappa_b\mathcal{D}[b]\rho, \qquad \mathcal{D}[c]\rho = c\rho c^\dagger - \tfrac{1}{2}\{c^\dagger c, \rho\}.$$

The supplied values are $g_2/2\pi = 0.763$ MHz, $\kappa_b/2\pi = 2.6$ MHz and $\kappa_a/2\pi = 9.3$ kHz, with storage amplitude $\alpha = 2.1$. Because $g_2/\kappa_b = 0.2935$ and $\epsilon_d/\kappa_b = 1.2942$, eliminating the buffer is not uniformly justified. Each simulation starts separately in $|+\alpha\rangle_a|0\rangle_b$ or $|-\alpha\rangle_a|0\rangle_b$.

For $A = \langle a\rangle$ and $B = \langle b\rangle$, factorization gives

$$\dot A = -2ig_2A^*B - i\epsilon_Z - \kappa_a A/2, \qquad \dot B = -i(g_2A^2 - \epsilon_d) - \kappa_b B/2.$$

Without memory loss or readout, $A = \pm\alpha$ and $B = 0$: the buffer drive and conversion amplitude cancel. The readout force first moves the memory in its imaginary direction and disturbs that cancellation. Linearize around $A = s\alpha + \delta A$, $s = \pm 1$. Neglecting loss and quadratic deviations, the settled memory equation is $0 = -2ig_2 s\alpha B_s - i\epsilon_Z$, hence

$$\boxed{B_s \simeq -s\frac{\epsilon_Z}{2\alpha g_2}, \qquad C_b \equiv |B_+ - B_-| \simeq \frac{|\epsilon_Z|}{\alpha g_2}.}$$

The buffer fields have opposite real signs. This weak-drive approximation applies after fast settling and before switching erases the initial-bit signal; the driven memory need not remain exactly at $\pm\alpha$.

Output distinguishability dephases a superposition, even without switching its bit. Unwanted switching is a separate process: drive-induced excitation can relax into the opposite logical branch.

Increasing the drive at fixed cat size

Here Γ_X is the logical population-imbalance decay exponent; a symmetric one-way switching rate is $\Gamma_X/2$. We estimate it from the Lindblad generator and check it against time evolution where decay is resolvable (Appendix B).

At $\alpha = 2.1$, the executed two-mode sweep gives:

ϵ_Z/g_2	Real-field contrast C_b	Γ_X (μs^{-1})
0.05	0.02379	6.12×10^{-6}
0.10	0.04757	2.37×10^{-5}
0.25	0.11885	1.62×10^{-4}
0.50	0.23689	9.24×10^{-4}
1.00	0.46037	8.61×10^{-3}

Over this sweep, a stronger drive produces a larger signal and a larger switching rate. These are static rates: they do not by themselves determine the error of a shorter measurement. A finite protocol must also be tested through its complete return and recovery.

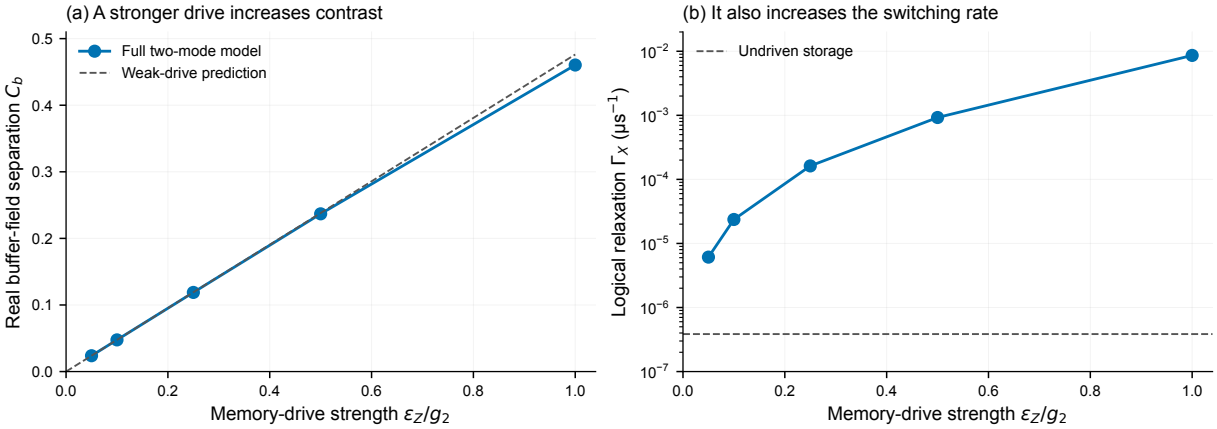


Figure 1: Fixed-size readout trade-off at $\alpha = 2.1$. Left: real buffer-field separation averaged over 3-4 µs, with the weak-drive prediction as a dashed guide. Right: estimated logical relaxation rate and the undriven-storage reference. Fields use $(32, 12)$ cutoffs, rate-sweep points $(28, 10)$, and the idle guide $(32, 12)$.

Task 2.4(c): propose and demonstrate one protocol

Objective and comparison

The fixed-size sweep suggests changing the protection during readout, rather than only increasing the force. Larger cat separation suppresses switching [2], but reduces the weak-drive response as $1/\alpha$. We therefore test a larger cat with a stronger drive: $\alpha = 2.6$ and $\epsilon_Z = 0.1g_2$.

The objective is a shorter complete cycle at a common signal target, with the estimated readout-stage Γ_X no larger than **undriven storage at** $\alpha = 2.1$. Speed is compared with the specified **weak readout at** $\alpha = 2.1$ **and** $\epsilon_Z = 0.05g_2$.

For the model's opposite real buffer responses, use ideal detection efficiency and the fixed real quadrature:

$$\mathcal{J}(T) = \kappa_b \int_0^T [\mathrm{Re}(B_+ - B_-)]^2 dt, \qquad \mathrm{SNR}^2 = 4\mathcal{J}.$$

This is a conditional-mean signal reference with unit white noise and optimal temporal weighting. Both protocols target $\mathcal{J} = 1$: a 15.9% ideal coherent-Gaussian classification error, not a demonstrated single-shot fidelity. The general measurement convention is in Appendix B.

Pulse sequence

Keep g_2 fixed and control cat size through the buffer microwave, $\epsilon_d(t) = g_2\alpha(t)^2$:

1. Enlarge $\alpha : 2.1 \to 2.6$ over 2 µs, with memory readout off.
2. Apply $\epsilon_Z = 0.1g_2$ and collect the buffer output for 41.3997 µs.
3. Turn off readout and restore $\alpha = 2.1$ over 2 µs.
4. Wait 2 µs, then evaluate the original storage logical states.

The two ramps use $s(u) = 10u^3 - 15u^4 + 6u^5$, $0 \le u \le 1$. The readout duration is chosen for the signal target; emission after that interval is excluded from the target. All ramps and recovery count in the complete-cycle time. The weak reference keeps $\alpha = 2.1$ throughout and has the same 2-µs recovery.

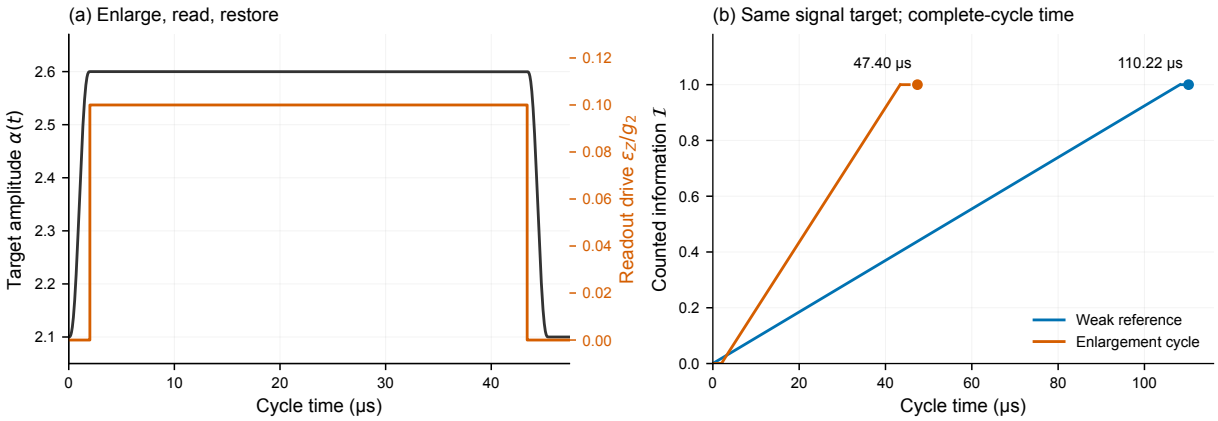


Figure 2: The selected controls and signal target. Left: the target cat amplitude (black, controlled by the buffer drive) and memory readout drive (orange, right axis). Right: counted information from saved trajectories. Dashed segments show return/recovery time after the measurement window; later emitted information is excluded. Markers give the complete-cycle durations.

Task 2.4(d): explain and validate the improvement

Complete-cycle results

We evaluate the memory in its original storage basis after recovery. The reported p_X is population in the opposite logical state, without postselection; leakage is population outside the orthonormal cat code space (Appendix B). These are directly integrated final populations, not a static rate multiplied by pulse duration.

Quantity	Weak reference	Enlargement cycle
Information at measurement end	1.0000000	0.9999925
Readout interval (µs)	108.2205	41.3997
Complete-cycle time (µs)	110.2205	47.3997
Final wrong-logical population p_X	3.3063×10^{-4}	7.9483×10^{-6}
Final code-space leakage	8.110×10^{-6}	8.110×10^{-6}

The enlargement cycle is **2.33 times faster at the same calculated signal target**, with a smaller recovered wrong-state population. This comparison includes its extra control time and tests whether the benefit survives return to the original cat size.

Why it works, and whether protection survives

Doubling ϵ_Z while increasing α from 2.1 to 2.6 predicts a field gain $2(2.1/2.6) = 1.615$. The quantum value is 1.616, and the information accumulation rate increases by 2.61. After both ramps and the common 2-µs recovery are included, the complete-cycle speed gain is 2.33.

The separate static-rate test gives $\Gamma_X = 2.0351 \times 10^{-7}\,\mu\text{s}^{-1}$ on the enlarged readout plateau, compared with 3.8475×10^{-7} for original undriven storage. The readout value is **0.529 times storage**, meeting the stated protection criterion. That rate test describes the plateau; the complete-cycle population test above also includes switching transients, the ramps and recovery. A separate equal-duration idle comparison is retained in Appendix B.

What the evidence supports

The observed gains exceed the tested cutoff changes. The last cycle-cutoff increment changes p_X by 0.775%; the last separate memory and buffer increments change the estimated static rate by 0.699% and 0.530%, with an earlier joint increment of 2.117%. These are empirical convergence checks, not rigorous error bars. Appendix B gives the full cutoff and independent-solver evidence.

The result is a demonstrated improvement over the specified weak reference **within the ideal cubic two-mode model**. It is not a global optimum over all controls. The circuit corrections in Appendix A have separate static and short-time checks, but the complete enlargement cycle has not been validated with them. Actual single-shot fidelity also remains untested. Additional pulse studies are supporting material in Appendix C.

Appendix A: circuit corrections

Off-resonant terms contribute at second pump order. For $H(t) = H_0 + \sum_{\Omega>0}(H_\Omega e^{i\Omega t} + H_\Omega^\dagger e^{-i\Omega t})$, the leading effective contribution is

$$\delta H = \sum_{\Omega>0} \frac{[H_\Omega, H_\Omega^\dagger]}{\Omega}.$$

Consistency through sixth order in zero-point phases requires both cubic-cubic and linear-fifth-order products. Keeping only cubic products would miss terms at the same formal order. Frequency shifts occur at fourth phase order; sixth order also produces $K_a n_a(n_a-1) + K_b n_b(n_b-1) + \chi n_a n_b$.

A separate sine-exchange model retains all zero-point phase orders in the selected resonant exchange channel, rather than every circuit resonance. Its buffer drive is recalibrated to cancel the coherent mean force. This does not restore exact coherent dark states. Corrected-model rates are compared with that same hardware's own idle reference.

Circuit calibration and separate numerical evidence

The retained sixth-phase-order effective coefficients before compensation are $K_a = -0.01754$, $K_b = -9.63496$ and $\chi = -1.43430\,\mu\text{s}^{-1}$. Their size makes an unqualified ideal-circuit extrapolation inappropriate. Constant frequency shifts and local Kerr slopes require retuning the memory, buffer and pump frequencies. The illustrative frequency shifts are approximately 0.922 and 22.87 MHz for memory and buffer.

A deliberate junction imbalance adds $C\cos\hat{\varphi}$ with $C = -2\Delta E_J J_0(\epsilon_p)$. Choosing

$$C = -\frac{\chi}{\varphi_a^2\varphi_b^2} = 4426.84\,\mu\text{s}^{-1}$$

cancels the retained cross-Kerr, requiring $\Delta E_J/E_J = -0.00881$ in this convention. It also changes both self-Kerrs; their retained residuals are approximately -0.00320 and $-0.67061\,\mu\text{s}^{-1}$. This is a concrete hardware/calibration proposal, not a free numerical cancellation.

With this compensation, static conservative-point rates in the phase-six model remain below that corrected hardware's own idle reference $8.39 \times 10^{-7}\,\mu\text{s}^{-1}$ over the tested cutoffs. However, the last separate memory and buffer variations are 2.49% and 4.69%, and there is no joint $(38, 12)$ point or corresponding idle buffer sweep. This is a cutoff-qualified feasibility witness, not a 1%-converged corrected-device certificate.

The earlier diagonal-only correction is retained as a restricted diagnostic. A separate full fundamental-pump van-Vleck matrix includes off-diagonal two-pump processes for finite-time signal sensitivity. Neither calculation supplies an exact Floquet open-system solution. No corrected full enlargement/read/restore cycle or fully corrected rare switching spectrum is delivered. Mixed pump-microwave counter-rotating terms, bath dressing, flux noise and fabrication uncertainty remain untested. Small spectral residuals cannot remove these physical model uncertainties.

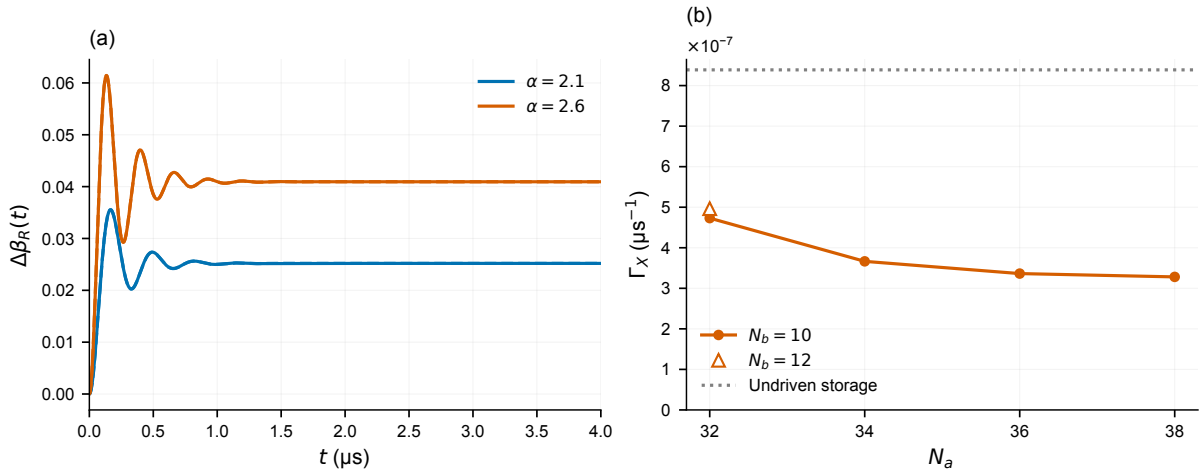


Figure 3: Circuit-model sensitivity after junction-imbalance compensation and frequency retuning. Left: phase-six and full fundamental-pump corrected field contrast. Right: phase-six selected relaxation rates versus memory cutoff, with a separate buffer increment and matched corrected idle guide. These are qualified partial-model results.

Appendix B: definitions and numerical validation

Logical populations and homodyne convention

Finite-α coherent states are not orthogonal. We use normalized even and odd cats $|C_\pm\rangle \propto |\alpha\rangle \pm |-\alpha\rangle$ and

$$P_{\text{code}} = |C_+\rangle\langle C_+| + |C_-\rangle\langle C_-|, \qquad |0_L, 1_L\rangle = (|C_+\rangle \pm |C_-\rangle)/\sqrt{2}.$$

For an initially positive codeword, $p_X = \langle 1_L|\rho_a|1_L\rangle$ after the stated recovery; the reverse branch is evaluated independently. We report the worse branch, without postselection. Leakage is $1 - \text{Tr}(P_{\text{code}}\rho_a)$. The initial coherent-state population in the opposite exact logical basis is 5.46×10^{-9} at α_0 and is retained. These populations are not counts of detected switching events.

For $B_s(t) = \langle b \rangle_s$ and $\Delta B = B_+ - B_-$, input-output theory gives $b_{\text{out}} = b_{\text{in}} + \sqrt{\kappa_b} b$. Under the unit-Wiener homodyne convention,

$$dY_s = 2\sqrt{\eta\kappa_b}\,\text{Re}(e^{-i\vartheta}B_s)dt + dW, \qquad \mathcal{J}_\vartheta(T) = \eta\kappa_b \int_0^T [\text{Re}(e^{-i\vartheta}\Delta B)]^2 dt.$$

Optimal temporal weighting then gives $\text{SNR}^2 = 4\mathcal{J}_\vartheta$. The local-oscillator phase ϑ is fixed during a record and optimized as stated; no time-dependent quadrature tracking is presumed. With coherent Gaussian outputs and equal priors, the ideal classification error is $\Phi_{\text{N}}(-\sqrt{\mathcal{J}})$, where Φ_{N} is the standard normal cumulative distribution. Thus $\mathcal{J} = 1$ **means 15.9% ideal Gaussian error, not 1% error**. The delivered $\mathcal{J}$ is a mean-signal shot-noise reference. Actual single-shot fidelity requires record statistics, including switching and non-Gaussian noise, which are not simulated here.

Rate extraction

Here Γ_X is the logical population-imbalance decay exponent used in Task 2.1. In a two-state model, $\Gamma_X = r_{01} + r_{10}$; each one-way rate equals $\Gamma_X/2$ only for symmetric switching. We identify a near-zero real Liouvillian mode that couples to the initial codeword

difference and $\mathrm{Re}\langle a\rangle$. Independent time-domain cases check the method where decay is resolvable. This avoids fitting the drive's initial displacement as a bit flip; it does not provide a full-spectrum proof that no other relevant slower mode exists.

Information-induced dephasing is a different effect

If the conditional emitted fields are coherent, their distinguishability causes an output-induced **coherence-decay exponent** $\gamma_{\mathrm{coh,out}} \simeq \kappa_b|\Delta B|^2/2$. This grows quadratically with weak readout strength even when neither logical branch switches. Under these assumptions it describes measurement dephasing, not Γ_X.

For an initial superposition, distinguishable output fields remove coherence even if neither logical branch switches. In fixed-size runs with $\kappa_a = 0$, normalized reduced-memory coherence at 4 µs is 1 without readout, 0.5346 at ratio 0.25, and 0.08565 at ratio 0.5. This attenuation includes residual entanglement with the buffer as well as information already emitted. It cannot all be identified with irreversible environmental dephasing at that instant.

Recovered wrong-logical populations in the pulse campaign are only of order 10^{-4}-10^{-3}. Setting $\kappa_a = 0$ reduces them by approximately 2-3%, leaving readout-induced population transfer. Thus strong loss of superposition coherence and much smaller unwanted switching can coexist. Temporary leakage, useful information gain, photon-loss phase errors and final logical transfer remain separate observables.

Weak-drive complex-field check

The leading response including memory loss is

$$B_s \simeq -s\frac{\epsilon_Z}{2\alpha g_2} + i\frac{\kappa_a}{4g_2}.$$

The imaginary offset is the same for both initial codewords and carries no bit contrast.

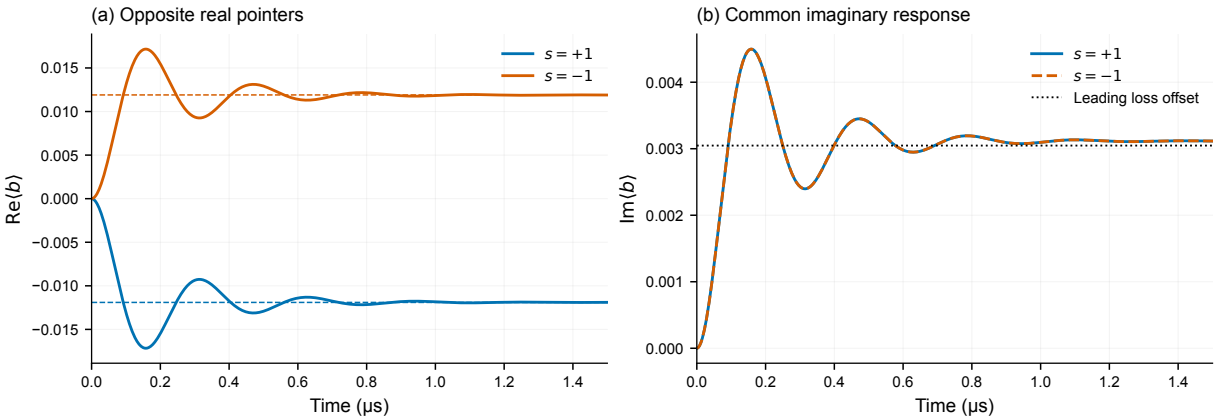


Figure 4: Weak-drive buffer response for both initial coherent codewords at $\alpha = 2.1$ and $\epsilon_Z/g_2 = 0.05$. Dashed real guides show the leading opposite pointers; the dotted imaginary guide shows the common photon-loss offset. The imaginary branch curves overlap.

Independent checks of the retained derivation

Direct laboratory-period averaging reproduces the fifth-order resonant Hamiltonian on a low-Fock block to $8.59 \times 10^{-13}\,\mu\mathrm{s}^{-1}$. A separate emitted-coherent-state overlap verifies the coherence-decay normalization. These check retained algebra, rather than the

physical adequacy of the approximation. The source and saved outputs are linked in the reference review.

Cutoff and independent-solver checks

All headline numbers are backed by saved JSON/NPZ outputs. Physical checks include Hermiticity, angular units, tensor ordering, ideal dark-state cancellation, exact logical projectors, trace conservation and upper-Fock populations. Numerical checks change memory and buffer cutoffs separately, tighten tolerances and time sampling, and compare independently implemented QuTiP and Dynamiqs solvers.

Evidence	Observed change / result	What it supports
Conservative rare rate, $(38, 16) \to (40, 16)$	0.699%	Last memory increment
Same rate, separate buffer increment	0.530%	Last buffer increment
Same rate, joint $(36, 14) \to (40, 16)$	2.117%	Earlier joint increment fails 1% target
Conservative cycle, $(36, 14) \to (38, 16)$	0.775% in final p_X	Empirical finite-cycle stability
Tighter integration of that cycle	5.1×10^{-14} absolute in p_X	Integration error below cutoff effect
Pulse campaign, same-cutoff independent solvers	Both selected pulses pass	Independent integration check
Campaign joint $(32, 12) \to (40, 14)$ plus backend change	About 0.0172% in final p_X	Supplemental cutoff/backend consistency

For the conservative rare mode, left/right eigen-residuals are 2.0×10^{-13} and $9.7 \times 10^{-11}\ \mu s^{-1}$; the observable loading is 5.197 versus the initial branch separation 5.2. Small algebraic residuals alone do not certify the infinite-Fock rate or identify the globally slowest relevant mode. The 0.699% increment is not a rigorous error bar. The optional fast point's static rate is less well converged, with about 4.6% last memory variation; its separate finite-cycle population check must not be substituted for rate convergence.

For the fixed-size pulses, the same-cutoff independent integrations compare all ten recorded observables on identical grids and pulse boundaries. Maximum wrong-population discrepancies are 1.0×10^{-13} for constant and 5.9×10^{-12} for raised cosine. The supplemental $(40, 14)$ Dynamiqs runs give final $p_X = 0.0018280291$ and 0.0015797447. Maximum all-time wrong-population differences from the original $(32, 12)$ runs are 3.63×10^{-7} and 3.19×10^{-7}. Trace errors stay below 5.2×10^{-15} and final minimum eigenvalues are approximately -6.5×10^{-9}, consistent with small integration negativity. These checks support the measured waveform differences, which are much larger than the observed numerical changes; they do not prove infinite-dimensional error bounds.

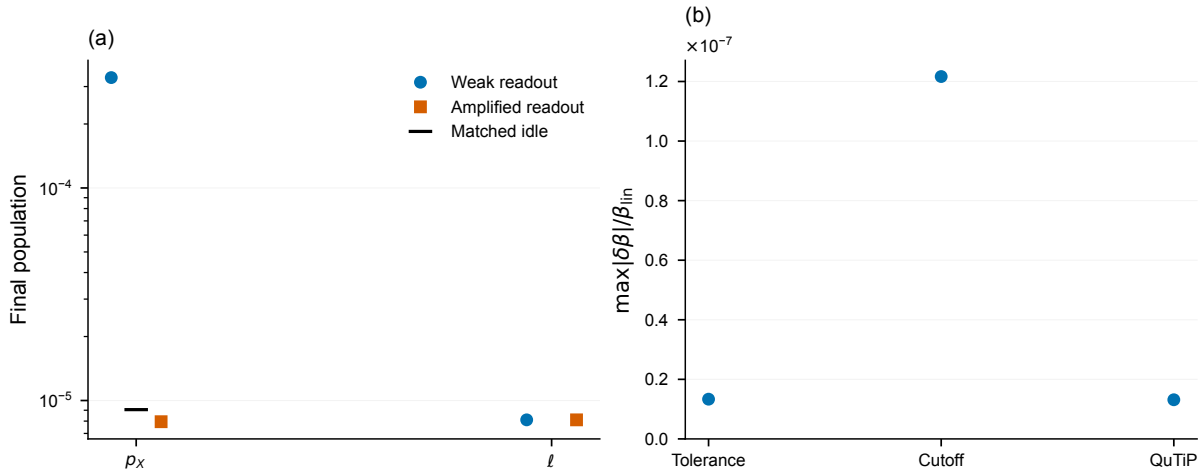


Figure 5: Curated numerical checks. Left: final wrong population and leakage; black mark: equal-clock idle at $(28, 10)$. Right: buffer-field differences normalized by the leading single-pointer amplitude, evaluated at common saved switch and final times. Tolerance and cutoff checks use the conservative cycle; QuTiP tests the fast cycle at the same $(32, 12)$ cutoff. Rate and complete-trajectory checks are reported separately.

Equal-duration idle check

An equal-duration undriven control gives $p_X = 9.0588 \times 10^{-6}$ over 47.3997 µs, versus 7.9483×10^{-6} for the selected cycle, a 12.3% lower population. The control uses $(28, 10)$ and the selected cycle $(38, 16)$; the control's static rate differs by 0.18% from $(32, 12)$. This is empirical support, not a same-cutoff inequality bound.

Robustness and interpretation of residual excitation

The fixed-size pulse campaign at $\alpha = 2.1$ varies drive amplitude, phase, exchange coupling and buffer damping. These are sensitivity tests of those pulses, not robustness certificates for the enlarged complete cycle. A ±5% drive-amplitude error changes information by approximately ±10% and wrong population by larger fractions; the Gaussian point ranges from 4.97×10^{-4} to 6.46×10^{-4} in recovered p_X. A ± 0.05 rad drive phase error splits the logical branches and increases its worst-branch population by about 9.75%. A reoptimized homodyne phase is a recalibrated measurement, so nominal- and optimized-quadrature results are retained separately.

Coupling errors at fixed buffer pump change the stabilized cat itself. The approximately 2.7×10^{-3} leakage observed under ±5% coupling mismatch is already present in matched zero-readout controls; attributing it all to measurement would be incorrect. Corrected circuit runs also break branch symmetry: for the constant example, the two recovered wrong populations are approximately 0.001676 and 0.002154. We report the worse branch rather than hiding asymmetry by averaging.

Enlarging the cat raises the leading photon-loss phase-dephasing scale by $2.6^2/2.1^2 = 1.53$ during readout. This is a cost in superposition coherence, even though the computational bit is better protected.

The enlargement-study cycles end near 1.75×10^{-5} buffer photons, not literal vacuum. Their excluded post-readout information is 0.000571, 0.001459 and 0.009148 for weak, conservative and fast cycles. This explicitly accounts for continued information leakage after switching off the memory drive. The small stationary buffer background is compatible with physical memory loss and does not constitute an exact reset proof.

A local stationary output-noise diagnostic is near the vacuum reference at the conservative operating point, but this does not establish white Gaussian noise across the full cycle. The report consequently makes no claim of experimentally achievable single-shot assignment fidelity.

Appendix C: additional controls and diagnostic checks

These calculations explain the selection of the main protocol and retain useful negative findings. They do not form part of the proposed enlargement/read/restore sequence.

Faster enlargement option

At $\alpha = 2.6$ and $\epsilon_Z/g_2 = 0.25$, 1-µs ramps and a 6.604-µs readout give a 10.604-µs cycle with $p_X = 1.2334 \times 10^{-5}$. Its static rate exceeds undriven storage and has about 4.6% last memory-cutoff variation. It is therefore not the selected storage-rate-preserving point.

Fixed-size pulse shaping

A separate frozen-source campaign explored constant, raised-cosine, Gaussian and build/hold/deplete envelopes, including signed termination segments and separate drive-phase perturbations. Semiclassical dynamics generated inexpensive candidates; full quantum evolution determined the results. A bounded differential-evolution search varied three interpretable parameters: peak ratio in $[0.05, 0.5]$, hold fraction in $[0.2, 1]$, and termination fraction in $[-1, 0]$. It maximized classical information minus a weighted end-buffer mean-field population, using several penalty weights, deterministic seeds and a six-iteration cap with population multiplier four. This was economical candidate generation, not an unconstrained or globally converged quantum optimization. Neither an optimizer's convergence nor a visually smooth pulse is evidence of scientific improvement.

For the three checked fixed-size pulses, the readout interval is 4 µs, recovery is 2 µs, peak ratio is 0.5, and the primary cutoff is $(N_a, N_b) = (32, 12)$:

Waveform	$\mathcal{J}$ in readout	Recovered p_X	Buffer photons at switch-off
Constant	3.6836	1.8277×10^{-3}	2.7065×10^{-2}
Raised cosine	3.2416	1.5795×10^{-3}	3.7637×10^{-3}
Gaussian	1.3080	5.6791×10^{-4}	9.918×10^{-5}

Raised-cosine shaping reduces end-buffer photons by 86.1%, recovered wrong population by 13.6%, and information by 12.0%. Gaussian shaping reduces the same quantities by 99.63%, 68.9% and 64.5%. These are **common-duration, common-peak comparisons**. They do not establish matched-information gains. Gaussian, rather than the proposed build/hold/deplete family, was selected by the low-residual-buffer criterion. The attractive depletion hypothesis therefore remains a trade-off, not a demonstrated universal winner.

An executed discovery pair illustrates the speed/error frontier more directly. Constant drives at ratio 0.5 for 2 µs and ratio 0.25 for 8 µs deliver nearly equal information, 1.84838 and 1.84741 (0.053% difference). Including recovery, their cycles are 4 and 10 µs, with final $p_X = 9.10 \times 10^{-4}$ and 6.56×10^{-4}. Faster acquisition costs about 38.7%

more wrong population. This pair used the discovery cutoff $(24, 8)$ and is not promoted as a final converged optimum.

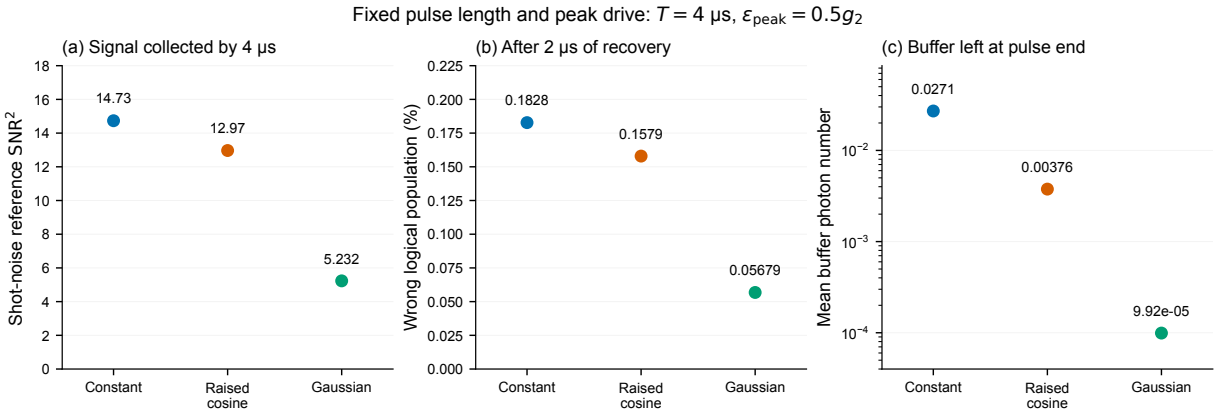


Figure 6: Executed waveform trade-offs at a common duration and peak drive. Information, recovered logical population and residual buffer excitation must be read together.

Why mean-field depletion failed the quantum test

For the same executed 2-μs candidate, classical information is 1.467 and quantum information is 1.423. Yet the predicted end-buffer population $|B(T)|^2 = 1.03 \times 10^{-4}$ is far below the quantum value $\langle n_b(T)\rangle = 3.61 \times 10^{-3}$. The reason is visible in the exact identity

$$\langle n_b\rangle = |\langle b\rangle|^2 + \langle (b - \langle b\rangle)^\dagger (b - \langle b\rangle)\rangle.$$

The quantum mean-field contribution is only 1.93×10^{-5}; fluctuations account for **99.47%** of the remaining photon population. Here fluctuations mean energy beyond the mean amplitude, not a claim of nonclassical photon statistics. The candidate was tested at discovery cutoff $(24, 8)$, so this is a demonstrated failure of its reset prediction rather than a precisely converged optimized protocol. A future depletion objective must constrain total photon population, as well as mean-field contrast and recovered logical error.

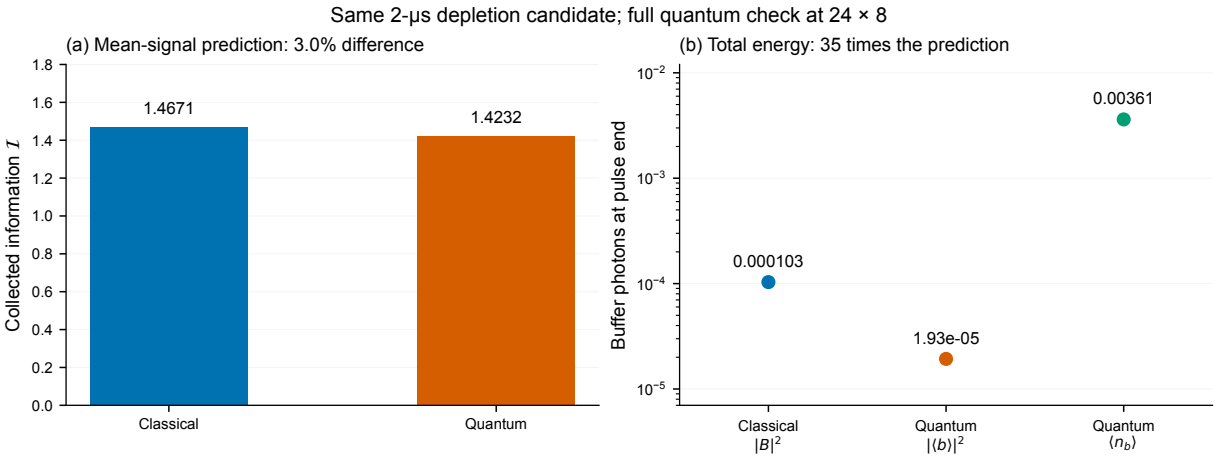


Figure 7: The same depletion candidate evaluated classically and quantum mechanically. Left: the integrated mean-signal prediction agrees within 3%. Right: actual photon population greatly exceeds both the classical prediction and the quantum mean-amplitude contribution. Readout lasts 2 μs; the quantum cutoff is $(24, 8)$.

Matched-information prefixes of a longer trajectory can locate information arrival, but do not simulate terminating that pulse and recovering. We therefore do not turn those

prefixes into unexecuted counterfactual error claims. A separate two-step input-shaping test shortened settling from 0.968 to 0.666 µs while reducing $\mathcal{J}(4\,\mu s)$ from 0.03705 to 0.03598. Suppressing ringing alone did not increase information over that fixed window.

Plateau information and population diagnostic

For additional interpretation, the rate and finite-pulse data can be connected: for symmetric rare switching during a settled plateau,

$$p_X(t) \simeq \frac{1 - e^{-\Gamma_X t}}{2}, \qquad \mathcal{J}(t) \simeq \dot{\mathcal{J}} t, \qquad q_X \equiv \frac{\Gamma_X}{2\dot{\mathcal{J}}}.$$

Thus q_X estimates switching probability per unit information in that limited regime. The supplied two-mode results give $q_X = 3.30 \times 10^{-4}$ for weak readout and 4.22×10^{-6} for enlarged readout: an approximately **78-fold lower plateau penalty per unit information**. The actual equal-information complete-cycle populations improve by **41.6-fold**, which is the relevant executed result; plateau estimates do not include the full transient, ramps and recovery.

This connects the independent efforts without treating them as identical calculations. For example, the local static rate at ratio 0.5 predicts $p_X(4\,\mu s) = 1.845 \times 10^{-3}$, while the cloud's recovered constant-pulse result is 1.828×10^{-3}, about 1% lower. This is a consistency diagnostic, using rate cutoff $(28, 10)$ and pulse cutoff $(32, 12)$, not a replacement for either convergence study or a general rate-to-pulse conversion.

Deliverables, reproduction and provenance

The dedicated Task 2.4 README is the single entry point for the final PDF, executed synthesis notebook, detailed calculations and exact reproduction commands. Notebook 10 rebuilds this synthesis from saved outputs; notebook 09 retains the detailed enlargement study. Modular source and simulation drivers remain separate from the frozen cloud implementation so that independent evidence is preserved.

The circuit and error mechanisms have prior treatments [2-4]; the present numerical results are indexed separately. The final evidence index records input hashes, exact derived metrics and source job identifiers. All 66 archived cloud source records match the execution manifest. The main campaign completed 104 of 106 jobs; its two historical joint-cutoff timeouts are preserved, and both supplemental checks passed separately. The final synthesis uses raw rate trajectories rather than the retained normalized legacy archive. PDF page renders and build checks accompany the report. Reconstructing the analysis requires no new quantum simulation.

AI-assistance disclosure

AI agents assisted with the circuit derivation, code, candidate design, numerical execution, independent audits and report synthesis. Independent implementations, saved trajectories, explicit convergence checks and adversarial review were used to test the results. This disclosure follows the challenge's allowance for AI-assisted code and explanation; it does not substitute for the submitting team's ability to explain and defend the work. No claim of human scientific approval is implied.

References

1. DuQuantum 2026 Alice & Bob challenge, supplied prompt v4, Task 2.4 and provided tutorials. Local source: `docs/challenge_prompt_v4.md`.
2. K. S. Dubovitskii, *Bit-flip errors in dissipative cat qubits: Second-order perturbation theory*, Physical Review A 111, 012617 (2025). Primary manuscript. Context for leakage-mediated errors; not a replacement for the present two-mode calculations.
3. U. Réglade et al., *Quantum control of a cat-qubit with bit-flip times exceeding ten seconds* (2024). Primary manuscript and supplementary derivations. Context for ATS pumping, semiclassical dynamics and calibration.
4. R. Gautier, M. Mirrahimi and A. Sarlette, *Designing High-Fidelity Zeno Gates for Dissipative Cat Qubits*, PRX Quantum 4, 040316 (2023). Primary manuscript. Prior analysis of buffer-mediated logical information and gate errors.